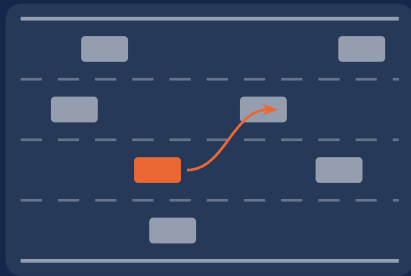


Neuro-Symbolic Safe Driving

A neural driver, real traffic law,
and three ways to join them

Silver Gjeka · VR527645

Traffic rules: Maierhofer et al., IEEE IV 2020 and 2022



INTRODUCTION

What We Will See

01

Learn to drive

three RL agents, pick the best

02

The rules

real traffic law from a paper

03

Apply the rules

three ways to make the car obey

04

A new world

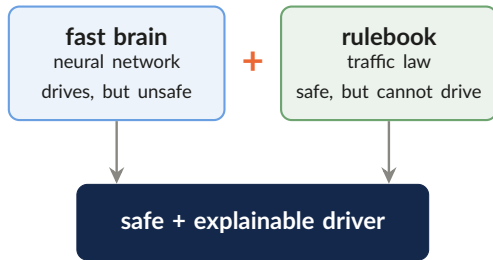
the same driver in a 3-D simulator

Every rule violation in this talk is counted by an **independent checker**: never by the reward or the shield the car uses. The numbers can be audited.

INTRODUCTION

The Goal: a Fast Brain and a Rulebook

- A **neural network** learns by itself to overtake traffic, but it drives dangerously and nobody can say *why* it acts.
- **Traffic rules** are written by humans: clear and checkable, but they cannot drive.
- **Neuro-Symbolic AI** joins the two: the network drives (System 1, fast), the rules check it (System 2, careful).
- The same rules then **explain** every decision: which rule, which fact, which step.

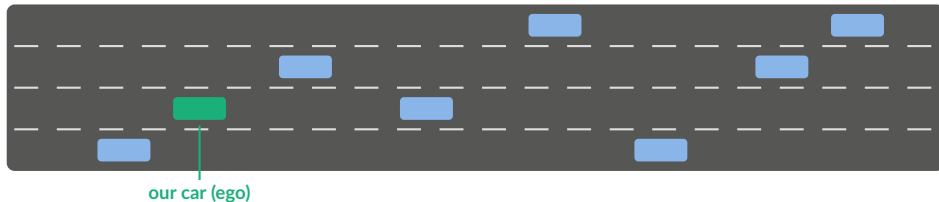


01

Learn to Drive

Three reinforcement-learning agents on a busy highway

The Highway Environment



Sees (numbers, not pixels)

itself and the 4 nearest cars:
present?, position, lane, speeds

Does (2 decisions / s)

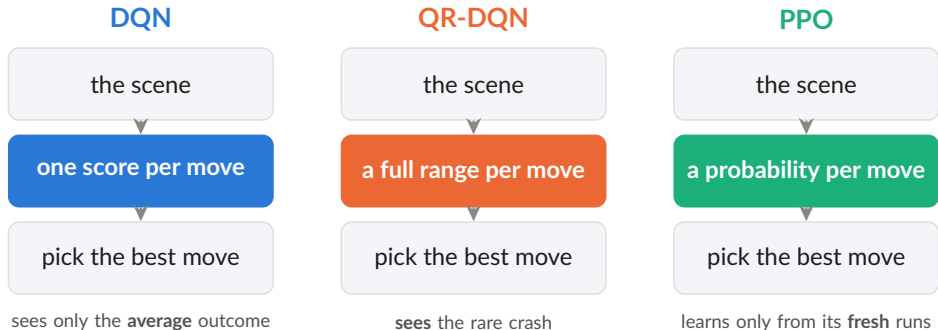
LANE_LEFT, LANE_RIGHT, FASTER,
SLOWER, IDLE

Reward (aggressive on purpose)

+ speed, + each car passed, – one
big penalty for a crash

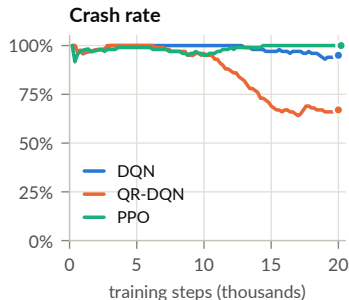
4 lanes, 50 cars, 40-second runs. The reward makes a **fast, risky** driver: exactly what the rules must fix later.

Three Agents, One Fair Test

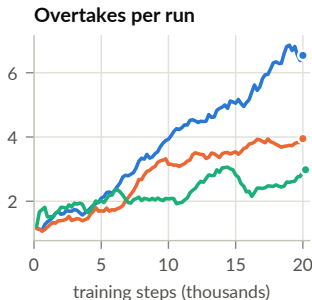


- **DQN** and **QR-DQN** are twins: QR-DQN only predicts 50 quantiles of the return instead of its mean.
- **Same for all:** road, traffic, reward, 20,000 training steps, and the same 50 test runs (10 seeds $\times$ 5).

Learning to Drive



DQN passes the most cars but never stops crashing: the best *average* is to drive flat out.

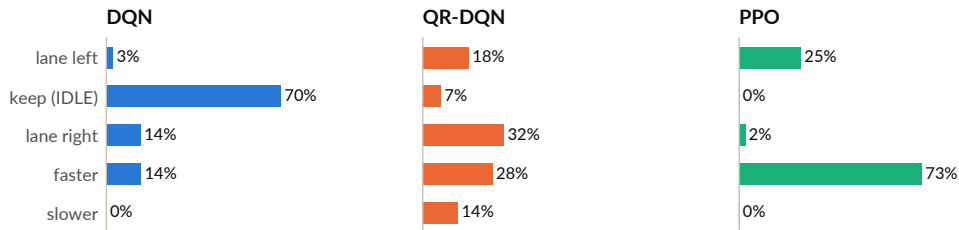


QR-DQN is the only one whose crashes fall and whose runs get longer: it keeps the rare crash in view.



PPO learns only from its own fresh runs: 20k steps are not enough.

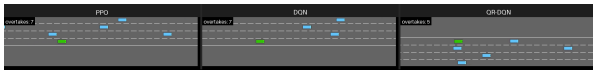
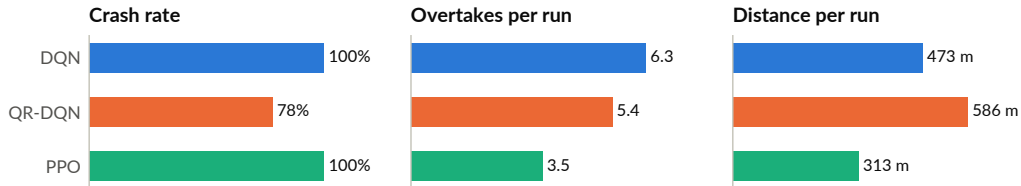
How the Three Drive



DQN mostly keeps its lane (70% IDLE), **PPO** mostly accelerates (73% FASTER). **QR-DQN** uses all five moves and is the **only one that brakes** (14% SLOWER): it crashes least.

Share of decisions over the same 50 test runs.

The Final Test: QR-DQN Wins



Best test run of each agent (videos/part1_three_agents.mp4)

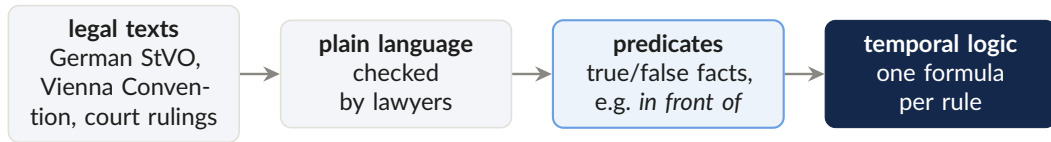
QR-DQN crashes least and drives furthest. 78% crashes is still a bad driver, on purpose: that is what the rules must fix.

02

The Rules

Real traffic law, written as logic

Where the Traffic Law Comes From



- **Metric temporal logic:** logic with time. G always, F eventually, P previously, O once in the past.
- Each rule becomes a formula a program can check. RG1, simplified:
$$G(\text{same lane} \wedge \text{in front} \wedge \neg \text{cut-in}) \Rightarrow \text{safe distance}$$

Tested on 2,500+ real German drivers

- Only about **37%** obey every rule.
- Most broken: **safe distance** (<65% comply) and **speed limit** (<78%).
- Almost nobody stops, brakes for no reason or passes on the right.

The Six Rules We Use

rule	in plain words	kind
RG1	keep a safe distance from the car ahead	HARD
RG3	do not go over the speed limit	HARD
RI1	do not stop in the middle of the road	HARD
RG2	do not brake harshly for no reason	SOFT
RG4	do not block the traffic behind you	SOFT
RI2	do not overtake on the right	SOFT

Hard: forbidden

Breaking one can directly cause a crash, so the shield **veto**es it.

Soft: discouraged

Bad habits that annoy or slow traffic, so the reward **penalises** them.

Safe distance (RG1), from the paper: my reaction and braking distance minus the leader's braking distance.

$$d_{\text{safe}} = v_{\text{ego}} t_d + \frac{v_{\text{ego}}^2}{2|a_{\text{ego}}|} - \frac{v_{\text{lead}}^2}{2|a_{\text{lead}}|}, \quad t_d = 0.3 \text{ s}, \quad a_{\text{ego}} = -10, \quad a_{\text{lead}} = -10.5 \text{ m/s}^2$$

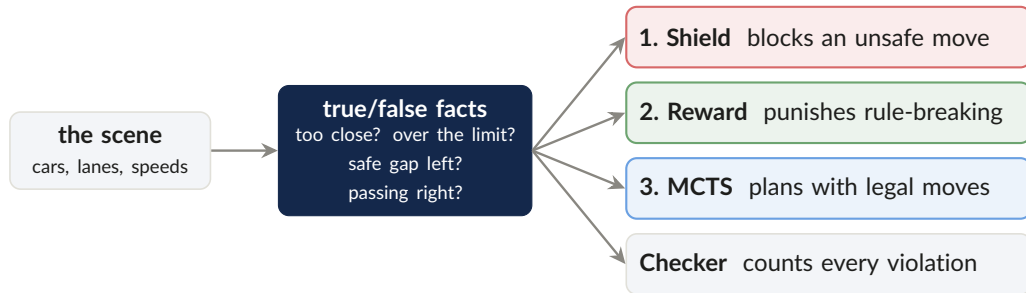
Thresholds: `configs/highway.yaml` (RG1 from the paper's Table II).

03

Apply the Rules

Shield, reward shaping and planning on the Part-1 winner (QR-DQN)

From the Scene to Facts, from Facts to Actions



The facts (**predicates**) are computed once per step and shared by all three methods *and* the checker: one source of truth.

Three Ways to Make the Car Obey

Shield (Lab 3)

A state machine checks each chosen move. If it breaks a **hard** rule, it is swapped for the safest legal one (brake or keep the lane).

- + no re-training
- cautious, fewer overtakes

Reward shaping

A short re-training where each **soft**-rule violation costs reward (right pass 0.5, blocking 0.3, harsh braking 0.3).

- + learns the habit
- no guarantee

MCTS planning (Lab 4)

Replaces the network: imagines 20 short futures (3 steps), 70% of moves drawn from the **legal** ones, and picks the best.

- + rules used *before* acting
- only as good as its model

Six setups on the same 50 test runs: baseline, +shield, +logic-reward, +shield+reward, +MCTS, +MCTS+shield.

PART 3 · APPLY THE RULES

Which Method Fixes Which Rule

	HARD rules · blocked by the shield			SOFT rules · penalised in the reward		
baseline (QR-DQN)	25%	59%	<1%	6%	27%	62%
+ shield	6%	16%	<1%	19%	33%	41%
+ logic-reward	35%	61%	<1%	5%	12%	37%
+ shield + reward	15%	13%	<1%	15%	32%	23%
+ MCTS	46%	96%	<1%	6%	1%	77%
+ MCTS + shield	40%	26%	<1%	32%	19%	42%
	RG1 safe distance	RG3 speed limit	RI1 no stopping	RG2 no harsh braking	RG4 keep the flow	RI2 no right pass

Share of steps that break each rule (darker = worse), same 50 test runs.

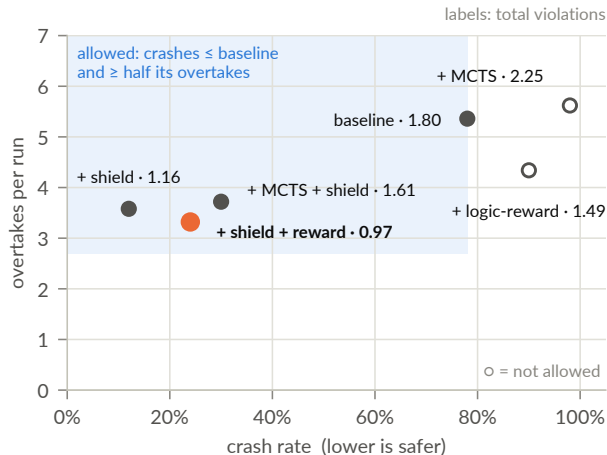
+**shield** crushes the **hard** rules, but brakes more.

+**reward** fixes the **soft** rules, but cannot stop speeding.

+**shield+reward**: both low, they fix **different** rules.

+**MCTS** plans for speed: it speeds and passes right the most.

Putting It Together: Shield + Reward Wins

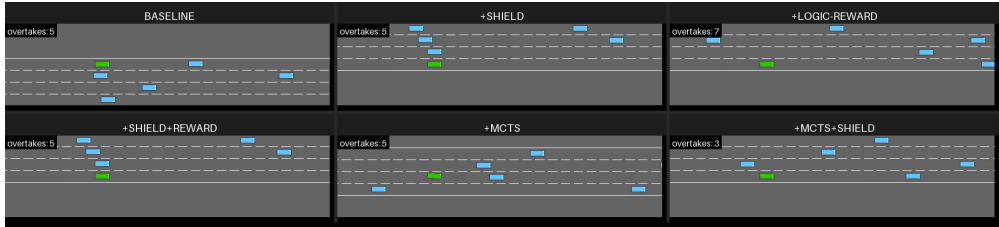


- **Allowed** (blue area): crashes no more than the baseline **and** at least half its overtakes.
- Among the allowed setups, the **fewest violations** wins: **shield + reward** (0.97, from 1.80).
- **Violations** = the six per-rule rates added up.

The shield alone is the safest: crashes **78% → 12%**, with no re-training at all.

PART 3 · APPLY THE RULES

Six Setups on the Same Road



Best test run of each setup (videos/part2_six_setups.mp4). Green = our car.

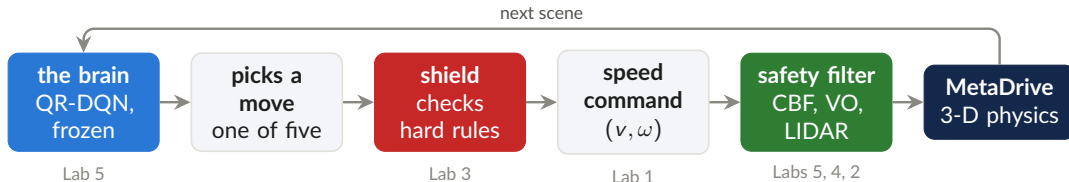
With the shield the car changes lane less and passes fewer cars, but it crashes far less.

04

A New World

The same driver and the same rules in MetaDrive, a 3-D robot simulator

The Same Driver, a New Simulator



Everything changes except the rules

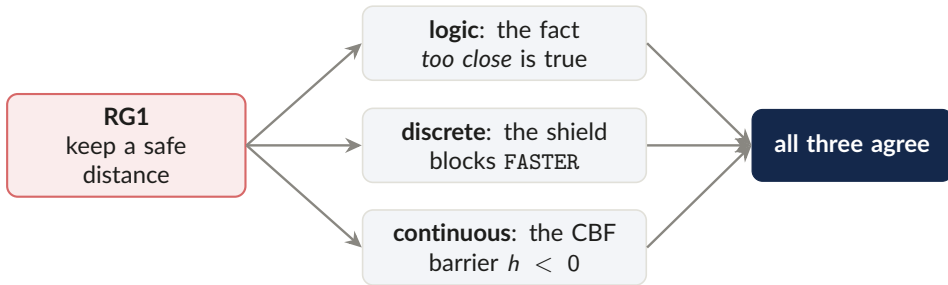
Other physics, 3 lanes instead of 4, starts from 0 m/s, robot-scale speeds (≈ 9 m/s). The network **never trained here**.

The safety filter

CBF keeps the gap and speed, **VO** blocks unsafe steering, **LIDAR** brakes for close cars in front.

Question: does the safety come from the **rules** or from the **network**? Three setups, 20 runs each on a ≈ 650 m road.

One Rule, Three Encodings



The same safe-distance rule, written three ways. A check (`rule_encoding_agreement`) confirms they **always agree**: the rule survived the move to a new world unchanged.

Same Rules, New World

	HARD rules · blocked by the shield			SOFT rules · penalised in the reward		
brain only	1%	0%	100%	0%	20%	66%
+ shield	<1%	0%	100%	0%	26%	48%
MCTS + shield	<1%	0%	100%	0%	28%	29%
	RG1 safe distance	RG3 speed limit	RI1 no stopping (threshold artifact)	RG2 no harsh braking	RG4 keep the flow	RI2 no right pass

Hard rules solved. The CBF enforces safe distance and speed at every step.

Dense traffic. A car often on the left means right passes (RI2) and slow flow (RG4); **MCTS** breaks the fewest.

RI1 is not a real failure. “Stopped” means below 18 m/s, set for the highway; the robot drives at ≈ 8 m/s.

Which Setup Wins Here?

35%

crash rate

brain only

2.15 passes per run

5%

crash rate

+ shield

only 1.40 passes per run

5%

crash rate

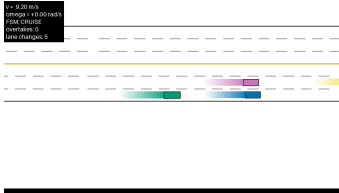
MCTS + shield

2.15 passes per run

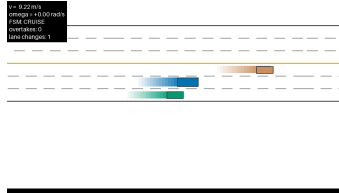
Why MCTS wins: the shield alone is safe but too cautious. MCTS plans on the **real** scene, so a new world does not confuse it, and it uses the rules **before** acting. The shield keeps it safe; MCTS gives back the overtaking.

PART 4 · A NEW WORLD

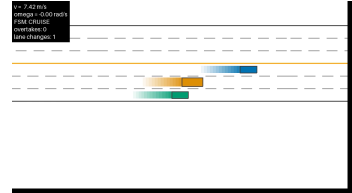
In Motion



brain only: crashes



+ shield: safe



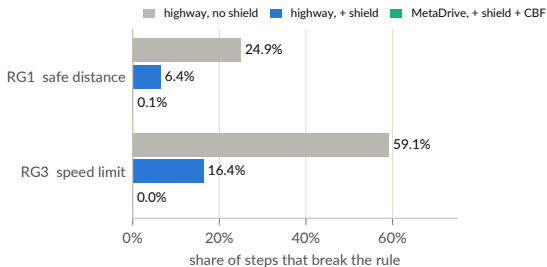
MCTS + shield: safe and passes



- Top: each setup's best run, from above. Green = our car.
- Left: the winner in the 3-D chase view; the panel shows the speed command (v, ω), the shield's state and the overtakes.
- All clips: [videos/](#) next to this file.

CONCLUSION

Hard Rules With the Shield: Lower, Not Zero

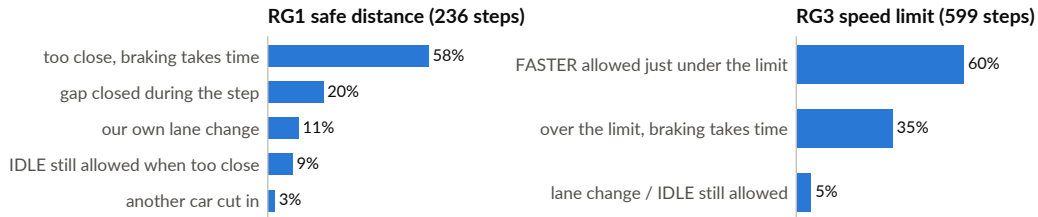


- On the highway the shield cuts **RG1** from 25% to 6.4% of steps and **RG3** from 59% to 16.4%, but not to zero.
- In MetaDrive, shield + **CBF** brings both to ≈ 0 .

To find out why, we **replayed the same 50 test runs** and logged, at every violating step, what the shield saw and what it did. The replay matches the saved results exactly.

CONCLUSION

Why: the Shield Is Reactive



Cause of every violating step in the replay of the +shield test runs.

Acts only after the fact. It brakes once the rule is broken; speed changes gradually (≈ 0.6 s), so the gap reopens slowly.

Checks now, not next. FASTER below 25 m/s sets the target to 30 m/s: the car overshoots.

Small gaps in the mask. IDLE allowed when too close. Cut-ins: only 3%.

Fix: a **predictive** shield that checks the state *after* the move, like the CBF in MetaDrive (≈ 0).

CONCLUSION

What We Learned

Lessons

- The network alone drives fast but crashes: **a reward is not enough.**
- A **shield** makes it safe with no re-training (78% → 12% crashes).
- A **shaped reward** teaches habits but gives no guarantee.
- Shield and reward fix **different** rules: together, the fewest violations.
- **MCTS + shield** was safe *and* overtaking in a world it never trained on.
- The same rules worked in **two simulators**, checked by an independent monitor.

Limits

- Short training (20k steps): the best driver still crashes in 78% of runs.
- Small tests: 50 runs on the highway, 20 in MetaDrive.
- Highway thresholds make RI1 and RI2 read too high on the robot.
- The highway shield is reactive; the CBF is a simple version (no QP).

CONCLUSION

What Comes Next: Intersection Rules

S. Maierhofer et al., **Formalization of Intersection Traffic Rules in Temporal Logic**, IEEE IV 2022

- The same method for junctions: traffic lights, stop and give-way signs, unregulated crossings.
- Stop at red lights and stop signs, give way to the right, yield to oncoming cars when turning left.
- Split into hard and soft rules, like the highway rules.

The plan: merge both papers into **one rule set** and switch rules with the place. The **same driver** then works safely in many worlds just by loading the right rules.

Thank you

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